

Double-Sided Iron-Core PMLSM Mover Teeth Arrangement Design for Reduction of Detent Force and Speed Ripple

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Abstract—This paper analytically and experimentally shows effectiveness and simplicity of mover teeth arrangement design, which dramatically reduces detent force and consequently speed ripple at low speed. For this purpose, this paper describes two design steps that include the conventional magnetic circuit optimization by response surface methodology (RSM) and a slight modification of the optimized model based on the proposed mover teeth arrangement. Two prototypes were fabricated, tested, and compared from various perspectives. It was experimentally confirmed that the proposed design is very effective in reduction of detent force and speed ripple with the least sacrifice of performance.

Index Terms—Detent force reduction, finite-element analysis (FEA), mover teeth arrangement, permanent-magnet linear synchronous motor (PMLSM), response surface methodology (RSM), speed ripple.

I. INTRODUCTION

A TRADITIONAL and still popular way to realize linear motion is rotary to linear conversion by combining a rotary motor and a mechanical conversion component such as rack-pinion and ball-screw. However, the conventional indirect approaches cause several problems which include conversion losses, structural complexity, backlash, maintenance, noise, vibration, and limitations on speed and travel distance. The other way is a direct drive by a linear motor, which removes all the mechanical conversion components and the problematic features mentioned above.

Driven by increasing and challenging demand for linear motion applications, the linear motors are becoming more attractive and desirable in many industries in line with advances in permanent-magnet (PM) materials, linear bearings, power electronics, and positioning sensors. For the past several decades, linear motors have been successfully adopted in a wide spectrum of application opportunities which include

transportation, machine tool, precision positioning mechanism, factory automation, semiconductor manufacturing equipment, and chip mounter [1].

The PMLSMs are classified into two types: ironless and iron-core PMLSMs according to the constructions of coil assembly. The ironless PMLSMs offer very smooth motion due to the absence of iron-core [2]–[4]. However, for many demanding linear applications which require high force density and efficient thermal dissipation, the iron-core PMLSMs have been emerged as the most suitable and dominant type of linear motor [5]–[18]. A typical construction of the iron-core PMLSM comprises a stator of PMs over which a mover translates and there are at least two problems especially caused in this particular topology: magnetic attraction and detent force. The magnetic attraction usually ranges from 5 to 10 times rated thrust, and it requires a set of heavy-duty linear bearings to maintain mechanical air-gap. For this reason, a double-sided iron-core PMLSM configuration is often preferred in the demanding linear applications, since the attraction can be mutually canceled out [14], [15]. In this configuration, the load on the linear bearing can be greatly reduced and the mechanical components can be simpler, and thus, the overall volume can also be reduced. However, whether it is single-sided or double-sided, the iron-core PMLSMs suffer the inherent detent force which is considered to be one of the most annoying problems in many servo applications, because the detent force further deteriorates positioning performance and motion smoothness especially at low speed [21].

Among the reduction techniques that were proposed in the previous researches, this paper specifically investigates the mover teeth arrangement [16]–[18] to show its comparative effectiveness and simplicity over the conventional geometry optimization approach. For this purpose, a double-sided iron-core PMLSM is optimized by response surface methodology (RSM) combined with 2-D finite-element analysis (FEA), and limitation of optimization is carefully discussed in comparison with the mover teeth arrangement design. The validity of the analysis and design is experimentally examined from various perspectives.

II. GENERAL PMLSM CHARACTERISTICS

A. Typical Speed and Thrust Profile

One of the popular applications of a short-stroke PMLSM is servo mechanism, which frequently requires precise linear motion. Therefore, force ripple needs to be reduced to a certain

Manuscript received July 2, 2015; revised October 12, 2015; accepted November 18, 2015. Date of publication December 22, 2015; date of current version April 8, 2016.

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Digital Object Identifier 10.1109/TIE.2015.2511079

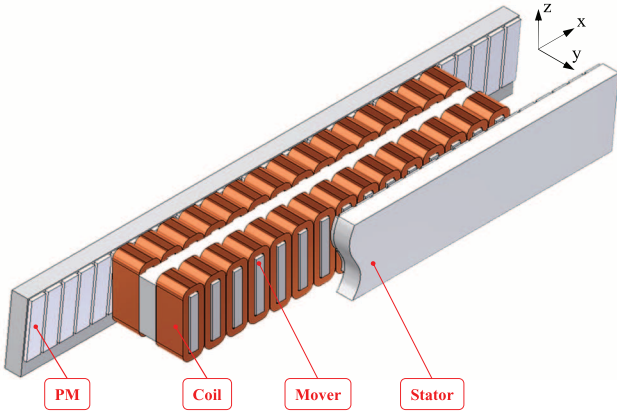


Fig. 1. Typical configuration of double-sided iron-core PMLSM.

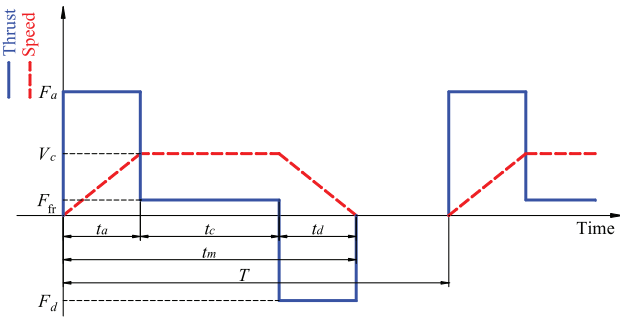


Fig. 2. Typical speed and thrust profile. F_a : acceleration thrust; F_{fr} : total friction; F_d : deceleration thrust; V_c : constant speed; t_a : acceleration time; t_c : constant speed time; t_d : deceleration time; t_m : total moving time; T : total cycle time.

level, since the ripple adversely affects motion smoothness and positioning control. In almost every PMLSM for this particular purpose, it is common to have a surface PM configuration where the PMs of alternating polarity are placed on stator surface, and there is no reluctance variation with respect to mover relative position. For this reason, detent force is the most dominant ripple origin, which needs to be effectively suppressed. Fig. 1 shows a typical configuration of double-sided iron-core PMLSM, which is discussed in this paper.

The importance of the detent force reduction can be best understood by examining a typical speed and thrust profile and friction components of a PMLSM. For an exemplary profile shown in Fig. 2, instantaneous thrust is frequently required during acceleration and deceleration, and the required thrust often exceeds 2–3 times rated thrust in these regions.

The motion can be expressed as (1) which shows that it only requires a small portion of rated thrust in the constant speed region, since the inertial force term is negligible and the amount of thrust required in that region is equivalent to frictional load

$$\begin{aligned} F_x &= m\ddot{x} + F_{fr} \\ F_{fr} &= c\dot{x} + \mu(F_y + mg_r) + F_{seal} \end{aligned} \quad (1)$$

where F_x , m , F_{fr} , c , μ , F_y , g_r , and F_{seal} denote thrust, mass of moving part, total friction, damping coefficient, friction coefficient, magnetic attraction, gravitational acceleration, and seal friction, respectively.

The total friction includes three major components: viscous friction, Coulomb friction, and constant friction, which mainly comes from linear bearing seals. The viscous friction plays an important role especially in high-speed operation, since it is linearly proportional to speed. The Coulomb friction is defined as the product of normal force and friction coefficient. The constant friction comes from linear bearing seals, and this component remains almost constant. For a typical double-sided iron-core PMLSM as shown in Fig. 1, magnetic attraction between mover and two stators are mutually canceled, and the Coulomb friction component is dependent mostly on the weight of the moving parts, and this component is usually small since typical friction coefficient of linear bearings range around 0.01 or often even less than that. Therefore, for this particular configuration, smaller size linear bearings can be used, and consequently, the total friction is comparatively lower than the single-sided iron-core PMLSM. If a double-sided iron-core PMLSM operates in the constant speed region especially at low speed, the total friction becomes close to the seal friction originating from the linear bearings, since the viscous friction has a negligible effect at low speed. In other words, detent force reduction is typically required for double-sided iron-core PMLSMs to reduce speed ripple at low speed.

B. Detent Force Components

Iron-core PMLSMs contain two major pulsating force components which include cogging force and end effect. The cogging force originates from magnetic interaction between PMs and slots, and this component can be easily reduced by skewing design, which is the most common reduction technique for iron-core PMLSMs. The end effect is caused by magnetic interaction between end teeth and PMs, and the period of this component is equal to pole-pitch (τ_p). The reduction of this component requires special design techniques, such as auxiliary teeth and mover teeth arrangement.

The previous research analytically investigated and discussed the end effect reduction, which provides physical interpretation as follows [7], [10], [22]:

$$F_R(x) = A_0 + \sum_{n=1}^{\infty} A_n \sin\left(\frac{2n\pi}{\tau_p}x + \alpha_n\right) \quad (2)$$

$$F_L(x) = -F_R(\tau_p - x - \delta) \quad (3)$$

$$\begin{aligned} F_{R+L}(x) &= F_R(x) + F_L(x) \\ &= \sum_{n=1}^{\infty} A_n \cos\left(\frac{\delta}{2}\right) \sin\left(\frac{2n\pi}{\tau_p}x + \frac{\delta}{2}\right) \end{aligned} \quad (4)$$

where F_R , F_L , F_{R+L} , A_0 , A_n , α_n , and δ denote force at right end, force at left end, resultant force at both ends, offset value, amplitude of n th harmonic component, phase angle of n th harmonic component, and phase difference between F_R and F_L , respectively.

The numerical representation (4) shows that the end effect can be effectively reduced by two different ways. One is minimization of the amplitude of the two forces, and this seems more than obvious. The other seems especially effective, since it utilizes the phase difference between the two forces, and the

TABLE III
DESIGN CANDIDATES

No.	W_p (mm)	H_p (mm)	W_t (mm)	Pk. detent force (N)	Cont. thrust (N)	Inst. thrust (N)	Mover volume (mL)	PM volume (mL)	Thrust/current density (N · mm ² /A)		Mover weight (g)		
									at cont. thrust	at inst. thrust	Core	Copper	Total
Global optimum	9.8	3.9	5.7	9.1	514	1201	4574	1.53	103	96	2328	2246	4574
1	12	2.0	3.0	2.3	452	761	4900	0.96	90	61	1829	3071	4900
2	12	3.2	5.6	7.5	549	1258	4698	1.54	110	101	2352	2346	4698
3	12	3.8	6.3	10.6	539	1290	4587	1.82	108	103	2457	2130	4587
4	12	2.5	5.8	5.8	546	1228	4803	1.20	109	98	2423	2380	4803
5	12	2.5	6.0	6.0	545	1237	4802	1.20	109	99	2467	2335	4802
6	12	2.5	6.5	6.6	538	1248	4798	1.20	108	100	2576	2222	4798

Global optimum is given by (6)–(11); No. 1 is given only by (6); No. 2 is given only by (7); No. 3 is given only by (8); No. 4 is given by (6)–(8); No. 5 is one of the practical design candidates; No. 6 is one of the practical design candidates.

In calculation of peak detent force and thrust for all the models, this paper applies $1/3\tau_p$ stator skewing using step skew modeling technique [23]. For the maximization of thrust, this paper considers two different current loading conditions to fully consider nonlinearity of thrust versus current characteristics, which comes from magnetic saturation at high current loading conditions. For the current loading conditions, this paper assumes a winding fill factor of 71.5%, a current density of $5A_{rms}/mm^2$ for the continuous loading since the model is close to totally enclosed condition and a current density of $12.5A_{rms}/mm^2$ for the instantaneous loading, which is 250% of the continuous loading. The mover adopts an open-slot topology, and prewound coils of high fill factor are inserted into the slots; however, it needs extra space within the slots for insulation, assembling, epoxy resin encapsulation, etc. Therefore, to consider the extra space, the winding fill factor is defined as (5), where A_c , H_s , and W_s denote total copper area, slot depth, and slot width, respectively.

For appropriate estimation of each response model, this paper adopts central composite design (CCD), whose arrays and corresponding calculated values are numerically listed in Table II. Statistical evaluation yields the corresponding responses which can be expressed as (6)–(11). It can be said that the response models are well established since R^2 values, which is coefficient of determination, of each response turn out to be as high as 0.9979, 0.9890, 0.9978, 1.0, 1.0, and 1.0, respectively,

$$\text{Winding fill factor} = \frac{A_c}{(H_s - 2) \times (W_s - 2)} \quad (5)$$

$$\begin{aligned} \text{Peak detent force} = & 1.2 + 1.07W_p - 1.87H_p \\ & - 2.267W_t - 0.097W_p^2 + 0.032H_p^2 + 0.034W_t^2 \\ & + 0.1451W_pH_p + 0.1477W_pW_t + 0.5214H_pW_t \end{aligned} \quad (6)$$

$$\begin{aligned} \text{Continuous thrust} = & -366 + 71W_p + 70.5H_p + 119.9W_t \\ & - 2.9W_p^2 - 7.17H_p^2 - 11.413W_t^2 - 0.59W_pH_p \\ & + 1.436W_pW_t - 3.294H_pW_t \end{aligned} \quad (7)$$

$$\begin{aligned} \text{Instantaneous thrust} = & -1092 + 40.3W_p + 309.9H_p \\ & + 428.7W_t - 1.19W_p^2 - 26.87H_p^2 - 32.65W_t^2 \\ & + 0.12W_pH_p + 3.72W_pW_t - 17.214H_pW_t \end{aligned} \quad (8)$$

$$\text{PM volume} = 0.04W_pH_p \quad (9)$$

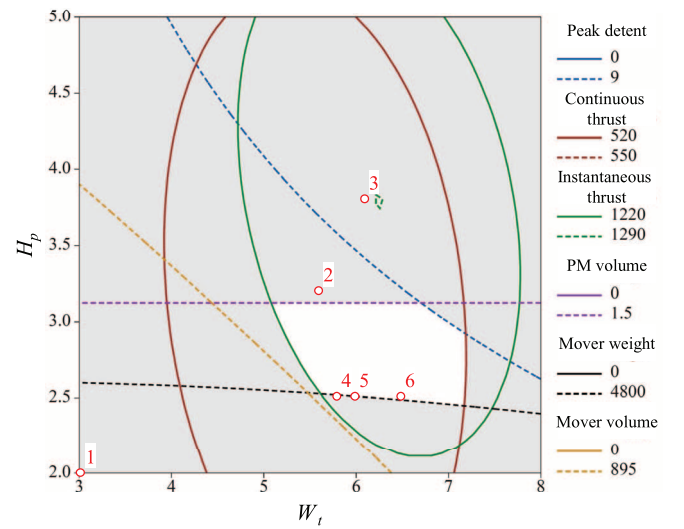


Fig. 4. Overlaid contour of responses.

$$\begin{aligned} \text{Mover weight} = & -5240.73 + 0.48W_p - 170.7H_p - 0.24W_t \\ & - 0.02W_p^2 - 0.02H_p^2 - 0.85W_t^2 - 1.17H_pW_t \end{aligned} \quad (10)$$

$$\begin{aligned} \text{Mover volume} = & -1067 - 31.86H_p - 18.09W_t \\ & - 0.54H_pW_t. \end{aligned} \quad (11)$$

There is only one set of design parameters that globally satisfies all the six responses at the same time. However, when examining each response separately, it can be seen that each response gives its own set of optimal values as compared in Table III. Considering the sets and the corresponding values, it can be fairly understood why a multiple-objective optimization is often considered to be a matter of choice after all, because even though the optimization of all the given objectives requires a reasonable compromise among the objectives. Therefore, it is a matter of designer's choice, and it is perfectly reasonable to examine the response surfaces by overlaying them together.

Fig. 4 shows the overlaid contour of the six responses when W_p is set as 12 mm, since all the three important responses related to the forces give the same optimal value for W_p as compared in Table III. In overlaying the contours, each low and high boundary is determined by individual optimization of the responses within about 5% sacrifice, which seems quite

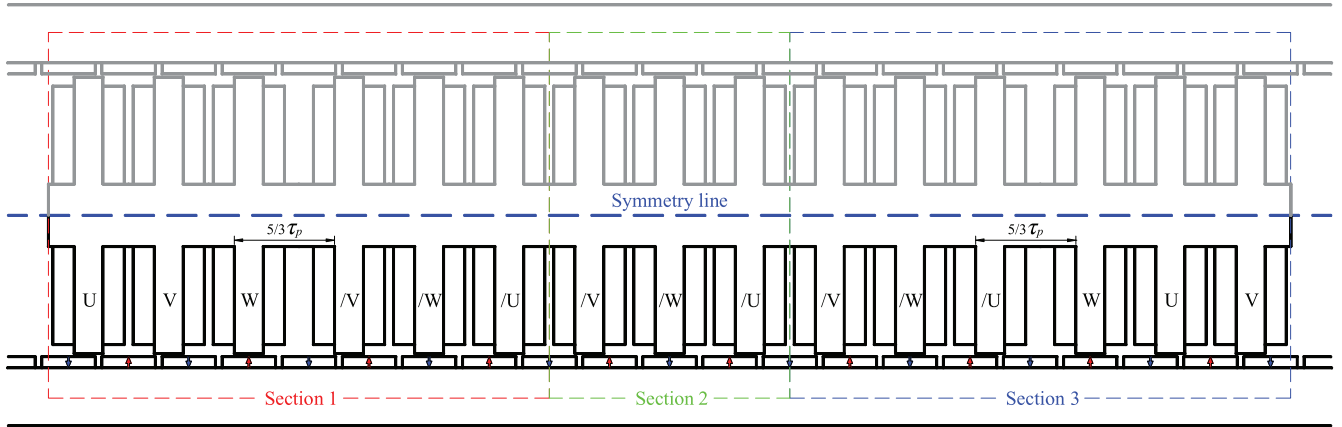


Fig. 5. Geometry of the proposed double-sided iron-core PMLSM.

acceptable and reasonable. For example, when drawing low and high boundaries for the continuous thrust, the optimal values obtained by (7) are chosen for the high boundary, and the low boundary is set as 5% lower than that. The same rule is applied for the other boundaries for the maximization objectives. For the minimization objectives, the high boundaries need to be bound and the global optimal values are referenced for setting the high boundaries within 5% compromise only if necessary.

Any set of design parameters that falls in the white area in Fig. 4 can satisfy the conditions mentioned above. It can be seen in Fig. 4 and Table III that a selection of design parameters neighboring No. 4 gives a reasonable set. However, when a realistic fabrication is taken into consideration, No. 5 or 6 can also be a practical candidate for prototyping and experimental verification, and both of them are quite comparable in every aspect. Considering all the values compared in Table III, this paper selects No. 6 as the optimal design for further characterization and comparison because of two practical reasons as follows: First, No. 6 shows better thrust versus current linearity and the higher instantaneous thrust characteristics than the other, i.e., No. 6 is more reasonable in the experimental point of view since it is less affected by magnetic saturation. Second, in mover material breakdown, No. 6 saves about 5% copper material, which is generally considered to be the second expensive material for motors. For these reasons, it can be said that No. 6 can be the practical choice in the manufacturing point of view. It should also be noted that No. 6 shows an even higher cost-effectiveness over No. 2 and 3 since No. 6 sacrifices only 2% in the continuous thrust, but saves 28% of PM material when compared to No. 2 and sacrifices only 3.3% in the instantaneous thrust when compared to No. 3 but still saves up to 52% of PM material.

The conventional multiple-objective optimization seems quite effective in the maximization of thrust at the cost of less than 5% sacrifice, and this approach even helps to select the most reasonable, practical, and cost-effective one when other objectives are carefully considered together. Taking a close look at Nos. 2, 3, and 4, which are obtained by individually or collectively considering the force-related responses, it can be observed that they are quite comparable in the continuous and the instantaneous thrust value within less than 5% difference.

Therefore, the average values of each one can serve as a reasonable estimation of the achievable optimal value ranges for the PMLSM shown in this paper under the given geometric constraints and loading conditions, and the estimation gives that the optimal continuous and instantaneous thrust would be somewhere around 540 N and 1260 N, respectively.

On the other hand, the achievable minimum peak detent force is found to be 2.3 N by No. 2; however, this comes far short of what it seems to be the optimal design since this choice requires excessive sacrifice and renders no practical use. It should be reminded that the geometry optimization by any means possible cannot reduce the peak detent force below to around 6 N within 5% compromise in other performance indices.

IV. MOVER TEETH ARRANGEMENT DESIGN AND COMPARATIVE ANALYSIS

A. Mover Teeth Arrangement Design

The conventional optimization approach without any special reduction design inevitably requires compromise as the best possible solution to conflicting design objectives with severe limitation in detent force reduction. For this reason, a couple of design techniques were introduced to effectively reduce the end effect of iron-core PMLSMs.

Literature survey revealed that there are at least two types of distinctive design techniques for end effect reduction: auxiliary teeth design [5]–[13] and mover teeth arrangement design [16]–[18]. The auxiliary teeth design is the most well-known technique, since the auxiliary teeth on both ends individually generates pulsating force components whose phase difference is 180° , and this technique can be applied for all iron-core PMLSMs regardless of number of mover teeth. However, this technique increases mover length and weight due to the auxiliary teeth on both ends, and also requires extensive geometry optimization, especially for the auxiliary teeth shapes and positions.

The mover teeth arrangement design is a relatively new technique which basically redesigns phase difference of teeth without any additional tooth elsewhere. The advantage that this technique offers for a three-phase machine is that it only

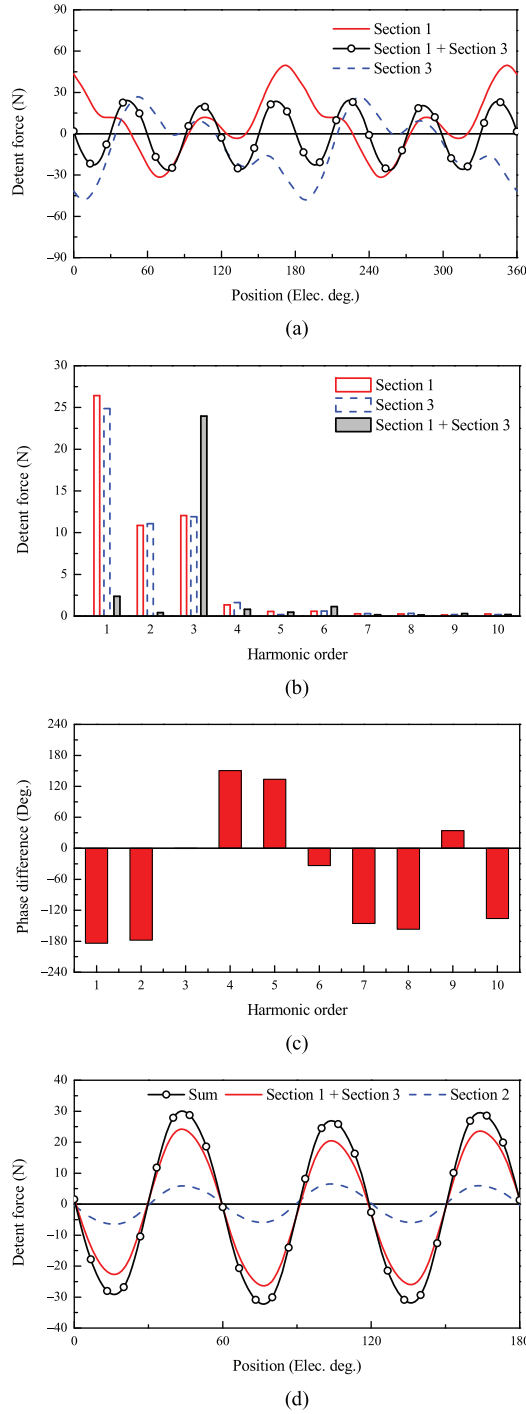


Fig. 6. Calculated detent forces. (a) Detent forces by sections 1 and 3. (b) Frequency components of detent forces by sections 1 and 3. (c) Phase differences of detent forces by sections 1 and 3. (d) Detent force waveforms by all three sections.

increases mover length by $1/3\tau_p$ or $2/3\tau_p$, which is a comparatively smaller increase in mover length and weight. On top of that, it only modifies relative positions of three or six mover teeth in the existing design without any performance sacrifice. However, this approach can only be available under such a condition that a PMLSM has at least six mover teeth and together with a proper skewing design to effectively reduce detent force.

It should be noted that the proposed design technique uses supplementary angle relationships to achieve the minimum

TABLE IV
PEAK DETENT FORCE COMPARISON

No.	W_p (mm)	H_p (mm)	W_t (mm)	Pk. detent force (N)		Reduction ratio (%)
				Conventional	Proposed	
1	9.0	2.0	3.0	2.30	0.67	70.9
2	12.0	2.0	3.0	2.10	0.70	66.7
3	9.0	5.0	3.0	6.50	3.17	51.2
4	12.0	5.0	3.0	6.50	3.67	43.5
5	9.0	2.0	8.0	5.10	1.47	71.2
6	12.0	2.0	8.0	6.10	1.39	77.2
7	9.0	5.0	8.0	16.1	7.26	54.9
8	12.0	5.0	8.0	19.4	8.33	57.1
9	9.0	3.5	5.5	7.20	2.73	62.1
10	12.0	3.5	5.5	8.30	3.02	63.6
11	10.5	2.0	5.5	3.90	0.90	76.9
12	10.5	5.0	5.5	12.1	5.31	56.1
13	10.5	3.5	3.0	4.30	2.07	51.9
14	10.5	3.5	8.0	12.0	4.58	61.8
15	10.5	3.5	5.5	7.90	2.96	62.5
						Average 62.0

separation span between the sections, and this particular technique can be further applicable for other multiphase machines, because the minimum separation span which is required to achieve the phase difference can be simply expressed as pole-pitch/ N_{ph} , and the minimum number of teeth is $2N_{ph}$, where N_{ph} denotes number of phase.

Fig. 5 shows the geometry of the proposed PMLSM design which is modified based on the previous design No. 6, and six teeth are rearranged, and consequently, the mover can be divided into three sections. In this configuration, it is intuitively perceivable that the section 2 only produces cogging force without end effect, since it is placed between the other two sections. The other two sections generate end effect and cogging force at the same time; however, fundamental components of the forces by the two sections are ideally identical but 180° shifted from each other due to the proposed teeth arrangement. This can be best understood by examining detent force waveforms produced by each section.

Fig. 6(a) specifically compares detent force waveforms produced by the two sections, and the resultant when skewing is not applied. It should be noted that the periods of the fundamental components of the waveforms by sections 1 and 3 are equally 180° which equals to τ_p , and the period of the fundamental component of the resultant waveform becomes 60° which corresponds to $1/3\tau_p$.

Fig. 6(b) compares frequency components of the waveforms, and it can be said that the forces produced by the sections 1 and 3 mainly consist of three components which include from the fundamental to the third harmonic. It should be carefully observed that the amplitudes of those components together with the corresponding phase differences shown in Fig. 6(c), since the fundamental and the second harmonic are dramatically reduced due to the phase differences. Consequently, the third harmonic becomes doubled due to nearly zero phase difference. Therefore, the resultant detent force contains only one dominant component, which is the third harmonic.

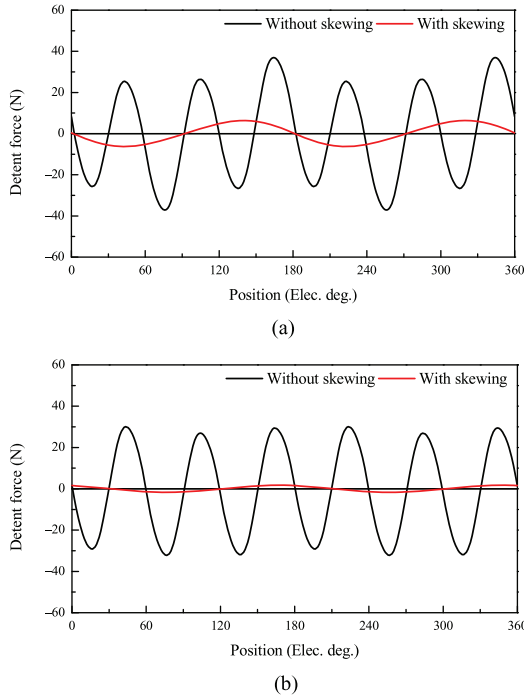


Fig. 7. Detent force waveform comparison. (a) Conventional design. (b) Proposed design.

Detent force waveforms by all three sections are also compared in Fig. 6(d), where it can be seen that section 2 only generates cogging force whose period is $1/3\tau_p$. It can also be observed that the sum of all detent forces becomes quite sinusoidal with a period of $1/3\tau_p$ as shown in Fig. 6(d). Therefore, the detent force of the proposed topology can be effectively reduced when $1/3\tau_p$ skewing is adopted either in mover or stator.

B. Comparative Analysis

This section investigates and discusses detent force characteristics of the conventional and proposed design to show the effectiveness of the proposed design. Table IV compares peak detent forces obtained for the models that are used in the previous optimization, and it can be seen that the proposed design reduces detent force by 62.0% on average.

Detent force waveforms of No. 6 and its modified model are specifically compared in Fig. 7, showing $1/3\tau_p$ skewing effects on each model at the same time. In the conventional design shown in Fig. 7(a), the detent force containing two dominant pulsating force components and $1/3\tau_p$ skewing suppresses only one component whose period is $1/3\tau_p$. However, in the proposed design shown in Fig. 7(b), the most dominant component, whose period is $1/3\tau_p$, can be effectively suppressed by $1/3\tau_p$ skewing. Therefore, the proposed design reduces the peak detent force by 73% (from 6.4 to 1.7 N) without any performance sacrifice.

V. EXPERIMENTS

Two prototypes are fabricated based on the analysis to confirm the validity of the design and analysis as shown in

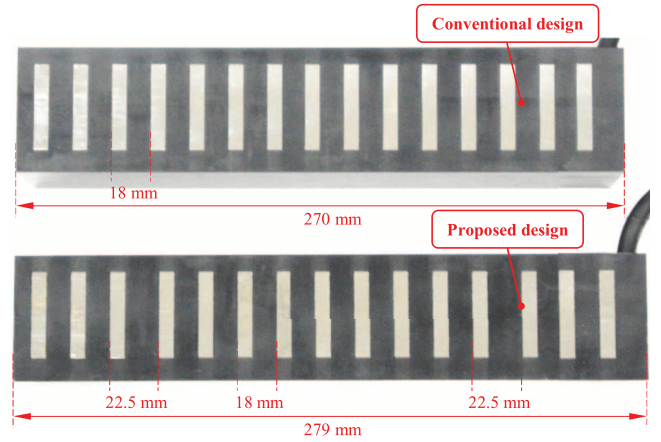


Fig. 8. Prototypes.

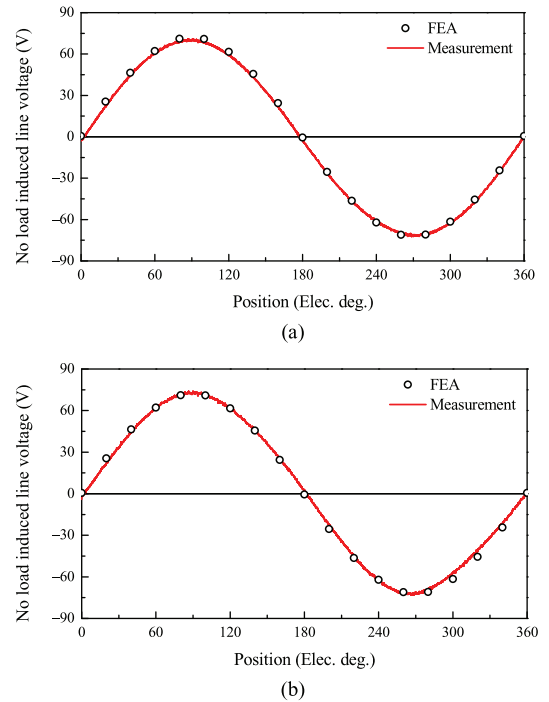


Fig. 9. No load-induced line voltage comparison at 1.0 m/s. (a) Conventional design. (b) Proposed design.

Fig. 8, which shows that the mover length of the proposed design increases by 9.0 mm, which corresponds to $2/3\tau_p$ when compared to the conventional design due to the teeth rearrangement.

To measure no load-induced line voltage at a constant speed, the test PMLSM is coupled to an ironless PMLSM and driven at the constant speed of 1.0 m/s, since the ironless PMLSM generates very smooth motion due to zero detent force. Fig. 9 shows no load-induced line voltage waveforms of the two prototypes measured at 1.0 m/s, showing that the analysis matches well with the measurements. It is also seen that the prototypes generate identical waveforms as previously mentioned, since the important magnetic design variables are exactly the same for both of them.

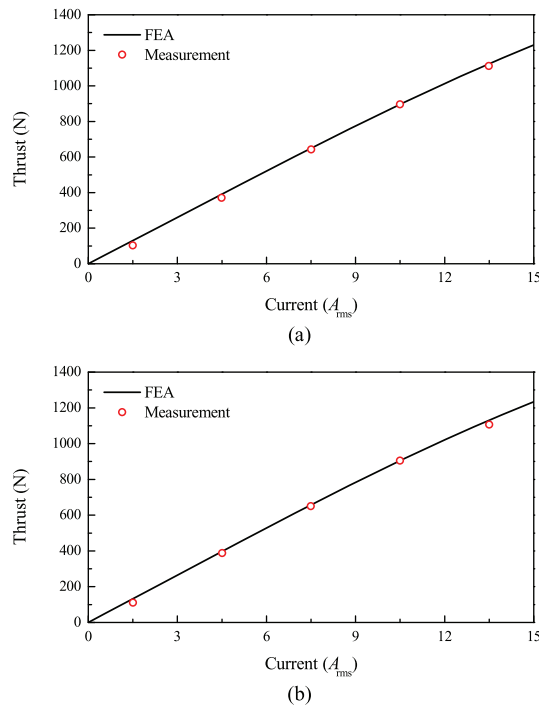


Fig. 10. Thrust versus current characteristics. (a) Conventional design. (b) Proposed design.

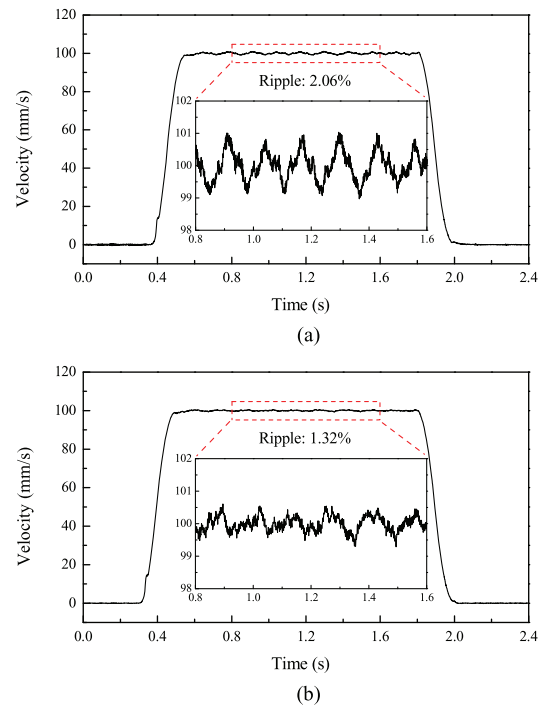


Fig. 12. Comparison of measured speed ripple at 100 mm/s. (a) Conventional design. (b) Proposed design.

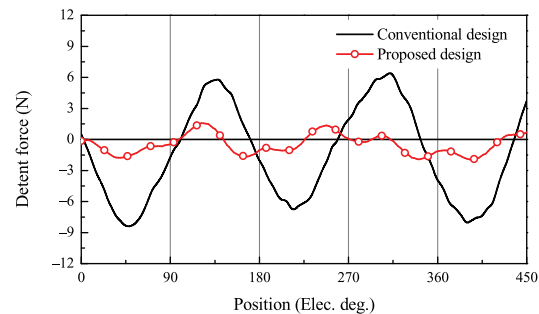


Fig. 11. Comparison of measured detent forces.

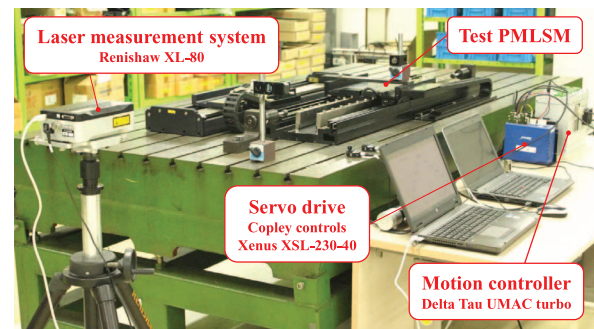


Fig. 13. Test setup for speed-ripple measurement.

Thrust can be measured with respect to mover position and applied current with a similar test setup described in the previous research [17], [18]. To obtain a reasonable average thrust at each loading conditions, the thrust was measured with respect to mover position over one electrical period to take thrust ripple into account. For the measurement of thrust under three-phase current excitation shown in Fig. 10, the two prototypes show exactly the same characteristics, showing a good agreement between the analysis and the measurement. The comparison shown in Figs. 9 and 10 truly confirms that the proposed design does not sacrifice any performance.

Like the no load-induced line voltage measurement, the test linear motor is coupled to the ironless PMLSM to minimize undesired ripple effect. To minimize viscous friction effect, the test PMLSM was driven by the ironless PMLSM at a constant speed of 10 mm/s. A force sensor is attached to a coupling component, which connects the two linear motors. However, the coupling component is configured only to push the moving part of the test linear motor due to the following reasons: The two

linear motors are not in perfect parallel to each other. If they are mechanically connected together, when the ironless PM linear motor moves the other, it leads to mechanical deformation most likely at the coupling component which has comparatively low stiffness, and the deformation eventually affects the measurement. This is also true for the thrust versus current measurement. Therefore, the thrust and the detent force measurements were carried out only in one direction due to this practical reason.

The measured detent forces of the two prototypes are compared in Fig. 11, which shows that the proposed design dramatically reduces detent force. For the comparison purpose of the two waveforms in one graph, the constant friction component, which mainly comes from the linear bearing seals, was excluded in this particular comparison. This can be done simply by subtracting the average detent force value over a couple of periods from the measurement data. However, when compared to the analysis, the measurements exhibit discrepancies which are possibly caused by several reasons such as magnetic leakage

effects that are not considered in 2-D FEA, fabrication imperfection, friction variation, and measuring sensor misalignment, etc.

Fig. 12 compares speed ripples at 100 mm/s measured with the test setup shown in Fig. 13. In this conservative comparison, the proposed design shows 36% lower speed ripple than the conventional design due to the reduced detent force.

VI. CONCLUSION

This paper has investigated and compared the two mover designs both analytically and experimentally. It is analytically discussed that the conventional optimization confronts a limitation on detent force reduction and inevitably requires a reasonable compromise; however, the proposed mover teeth arrangement effectively reduces detent force by 73% without any notable sacrifice in performance. It was experimentally shown that the proposed design also reduces speed ripple at low speed by 36%.

It is highly expected that the proposed design can be successfully implemented in various iron-core PMLSMs with the least changes.

REFERENCES

- [1] M. Y. Chen and J. S. Lu, "High-precision motion control for a linear permanent magnet iron core synchronous motor drive in position platform," *IEEE Trans. Ind. Informat.*, vol. 10, no. 1, pp. 99–108, Feb. 2014.
- [2] L. Li, D. Pan, and X. Huang, "Analysis and optimization of ironless permanent-magnet linear motor for improving thrust," *IEEE Trans. Plasma Sci.*, vol. 41, no. 5, pp. 1188–1192, May 2013.
- [3] S. I. Kim, J. P. Hong, Y. K. Kim, H. Nam, and H. I. Cho, "Optimal design of slotless-type PMLSM considering multiple responses by response surface methodology," *IEEE Trans. Magn.*, vol. 42, no. 4, pp. 1219–1222, Apr. 2006.
- [4] Y. Zhang, Z. Yang, M. Yu, K. Lu, Y. Ye, and X. Liu, "Analysis and design of double-sided air core linear servo motor with trapezoidal permanent magnets," *IEEE Trans. Magn.*, vol. 47, no. 10, pp. 3236–3239, Oct. 2011.
- [5] S. W. Youn, J. J. Lee, H. S. Yoon, and C. S. Koh, "A new cogging-free permanent-magnet linear motor," *IEEE Trans. Magn.*, vol. 44, no. 7, pp. 1785–1790, Jul. 2008.
- [6] Y. W. Zhu, D. H. Koo, and Y. H. Cho, "Detent force minimization of permanent magnet linear synchronous motor by means of two different methods," *IEEE Trans. Magn.*, vol. 44, no. 11, pp. 4345–4348, Nov. 2008.
- [7] Y. W. Zhu, S. G. Lee, K. S. Chung, and Y. H. Cho, "Investigation of auxiliary poles design criteria on reduction of end effect of detent force for PMLSM," *IEEE Trans. Magn.*, vol. 45, no. 6, pp. 2863–2866, Jun. 2009.
- [8] C. C. Hwang, P. L. Li, and C. T. Liu, "Optimal design of a permanent magnet linear synchronous motor with low cogging force," *IEEE Trans. Magn.*, vol. 48, no. 2, pp. 1039–1042, Feb. 2012.
- [9] K. B. Jang, J. H. Kim, H. J. An, and G. T. Kim, "Optimal design of auxiliary teeth to minimize unbalanced phase due to end effect of PMLSM," *IEEE Trans. Magn.*, vol. 47, no. 5, pp. 1010–1013, May 2011.
- [10] N. Baatar, H. S. Yoon, M. T. Pham, P. S. Shin, and C. S. Koh, "Shape optimal design of a 9-pole 10-slot PMLSM for detent force reduction using adaptive response surface method," *IEEE Trans. Magn.*, vol. 45, no. 10, pp. 4562–4565, Oct. 2009.
- [11] R. Cao, M. Cheng, C. C. Mi, and W. Hua, "Influence of leading design parameters on the force performance of a complementary and modular linear flux-switching permanent-magnet motor," *IEEE Trans. Ind. Electron.*, vol. 61, no. 5, pp. 2165–2175, May 2014.
- [12] R. Cao, M. Cheng, and W. Hua, "Investigation and general design principle of a new series of complementary and modular linear FSPM motors," *IEEE Trans. Ind. Electron.*, vol. 60, no. 12, pp. 5436–5446, Dec. 2013.
- [13] H. Zhang, B. Kou, Y. Jin, and H. Zhang, "Investigation of auxiliary poles optimal design on reduction of end effect detent force for PMLSM with typical slot-pole combinations," *IEEE Trans. Magn.*, vol. 51, no. 11, p. 8203904, Nov. 2015.
- [14] B. Q. Kou, H. X. Wu, L. Y. Li, L. L. Zhang, Z. Zhao, and H. C. Cao, "The thrust characteristics investigation of double-Side plate permanent magnet linear synchronous motor for EML," *IEEE Trans. Magn.*, vol. 45, no. 1, pp. 501–505, Jan. 2009.
- [15] S. G. Lee, S. A. Kim, S. Saha, Y. W. Zhu, and Y. H. Cho, "Optimal structure design for minimizing detent force of PMLSM for a ropeless elevator," *IEEE Trans. Magn.*, vol. 50, no. 1, p. 4001104, Jan. 2014.
- [16] S. U. Chung *et al.*, "A feasibility study on a new doubly salient permanent magnet linear synchronous machine," *IEEE Trans. Magn.*, vol. 46, no. 6, pp. 1572–1575, Jun. 2010.
- [17] S. U. Chung, J. M. Kim, B. C. Woo, D. K. Hong, and J. Y. Lee, "Development of doubly salient permanent magnet linear synchronous motor for general-purpose automation applications," *Int. J. Precis. Eng. Manuf.*, vol. 14, no. 12, pp. 2075–2080, Dec. 2013.
- [18] S. U. Chung *et al.*, "Design and experimental validation of doubly salient permanent magnet linear synchronous motor for precision position control," *Mechatronics*, vol. 23, no. 2, pp. 172–181, Mar. 2013.
- [19] X. K. Gao, T. S. Low, Z. J. Liu, and S. X. Chen, "Robust design for torque optimization using response surface methodology," *IEEE Trans. Magn.*, vol. 38, no. 2, pp. 1141–1144, Mar. 2002.
- [20] H. M. Hasanien, A. S. Abd-Rabou, and S. M. Sakr, "Design optimization of transverse flux linear motor for weight reduction and performance improvement using response surface methodology and genetic algorithms," *IEEE Trans. Energy Convers.*, vol. 25, no. 3, pp. 598–605, Sep. 2010.
- [21] M. Wang, L. Li, and D. Pan, "Detent force compensation for PMLSM systems based on structural design and control method combination," *IEEE Trans. Ind. Electron.*, vol. 62, no. 11, pp. 6845–6854, Nov. 2015.
- [22] Z. Q. Zhu, Z. P. Xia, D. Howe, and P. H. Mellor, "Reduction of cogging force in slotless linear permanent magnet motors," *Proc. Inst. Elect. Eng.—Elect. Power Appl.*, vol. 144, no. 4, pp. 277–282, Jul. 1999.
- [23] S. Williamson, T. J. Flack, and F. Volschenk, "Representation of skew in time-stepped two-dimensional finite element models of electrical machines," *IEEE Trans. Ind. Appl.*, vol. 31, no. 5, pp. 1009–1015, Sep./Oct. 1995.



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